

Linear Algebra HW 6

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16th February 2026

Question 1

If the image of a linear transformation is equal to the kernel of the same linear transformation, then by the rank-nullity theorem we can see that $\dim(\text{Im } T) = \dim(\ker T) = 2$

i.e. we need to send a two dimensional subspace to $\vec{0}$ ($\ker T$) and then the remaining two dimensional needs to be mapped to the $\ker T$

One such transformation is $T : \mathbb{R}^4 \rightarrow \mathbb{R}^4$, $T(\vec{v}) = A\vec{v} \quad \forall \vec{v} \in \mathbb{R}^4$

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

As the first two basis vectors get mapped to $\vec{0}$ and the other two basis vectors get mapped to the first two basis vectors

Question 2

Let $\vec{v}_1, \vec{v}_2 \in V$ and $\alpha \in \mathbb{R}$,

- $(S + T)(\vec{v}_1 + \vec{v}_2) = S(\vec{v}_1 + \vec{v}_2) + T(\vec{v}_1 + \vec{v}_2) = S\vec{v}_1 + T\vec{v}_1 + S\vec{v}_2 + T\vec{v}_2 = (S + T)(\vec{v}_1) + (S + T)(\vec{v}_2)$
- $(S + T)(\alpha\vec{v}_1) = S(\alpha\vec{v}_1) + T(\alpha\vec{v}_1) = \alpha(S(\vec{v}_1) + T(\vec{v}_1))$
- Thus, $(S + T)$ is a linear transformation
- $(\lambda T)(\vec{v}_1 + \vec{v}_2) = \lambda((T)(\vec{v}_1 + \vec{v}_2)) = \lambda(T(\vec{v}_1) + T(\vec{v}_2)) = \lambda T(\vec{v}_1) + \lambda T(\vec{v}_2)$
- $(\lambda T)(\alpha\vec{v}_1) = \lambda(T(\alpha\vec{v}_1)) = \lambda\alpha T(\vec{v}_1) = \alpha(\lambda T(\vec{v}_1))$
- Thus (λT) is a linear transformation

Question 3

A diagonalizable $\iff \exists$ a basis of $\mathbb{R}^3$ consisting of eigenvectors of $A \iff A$ has n distinct eigenvalues
To first find the eigenvalues we solve the characteristic polynomial of A equal to zero

$$\chi_A(\lambda) = \det(A - \lambda I) = 0$$

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 6 - \lambda & -10 & 1 \\ 1 & -11 - \lambda & 3 \\ 4 & -40 & 11 - \lambda \end{vmatrix} \\ &= (6 - \lambda)((-11 - \lambda)(11 - \lambda) + 120) + (10)((11 - \lambda) - 12) + (-40 + 4(11 + \lambda)) \\ &= (6 - \lambda)(\lambda^2 - 1) - 10(\lambda + 1) + 4 + 4\lambda \\ &= (6 - \lambda)(\lambda + 1)(\lambda - 1) - 10(\lambda + 1) + 4(\lambda + 1) \\ &= (\lambda + 1)(6\lambda - 6 - \lambda^2 + \lambda - 10 + 4) \\ &= -(\lambda + 1)(\lambda - 3)(\lambda - 4) \end{aligned}$$

We have three distinct eigenvalues $\lambda = -1, \lambda = 3$ and $\lambda = 4$,
Hence A is diagonalizable

To find the diagonal matrix D to which A is similar we note that in the basis of $\mathbb{R}^3$ consisting of eigenvectors, say $\mathcal{B} = \{\vec{b}_1, \vec{b}_2, \vec{b}_3\}$, that each of them is only scaled, hence $[A]_{\mathcal{B}, \mathcal{B}}$ is diagonal.

$$[A]_{\mathcal{B}, \mathcal{B}} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Thus $D = B^{-1}AB$ where $D = [A]_{\mathcal{B}, \mathcal{B}}$ and $B = \mathcal{P}_{\mathcal{B} \rightarrow \mathcal{E}}$

To find the eigenvectors we look in $\mathcal{N}(A - \lambda I)$,

$$\begin{aligned} \mathcal{N}(A + I), & \begin{bmatrix} 7 & -10 & 1 & | & 0 \\ 1 & -10 & 3 & | & 0 \\ 4 & -40 & 12 & | & 0 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 0 & -1 & | & 0 \\ 0 & 3 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \end{aligned}$$

One such eigenvector is $\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$

$$\begin{aligned} \mathcal{N}(A - 3I), & \begin{bmatrix} 3 & -10 & 1 & | & 0 \\ 1 & -14 & 3 & | & 0 \\ 4 & -40 & 8 & | & 0 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 0 & -2 & | & 0 \\ 0 & 4 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \end{aligned}$$

One such eigenvector is $\begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$

$$\begin{aligned} \mathcal{N}(A - 4I), & \begin{bmatrix} 2 & -10 & 1 & | & 0 \\ 1 & -15 & 3 & | & 0 \\ 4 & -40 & 7 & | & 0 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 0 & -3 & | & 0 \\ 0 & 4 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \end{aligned}$$

One such eigenvector is $\begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$

Thus,

$$B = \begin{bmatrix} \vec{b}_1 & \vec{b}_2 & \vec{b}_3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 3 & 4 & 4 \end{bmatrix}$$

Question 4

We repeat the process of the last question,

$$\begin{aligned} \chi_A(\lambda) &= \det \begin{bmatrix} -5\lambda & 0 & 4 \\ -11 & 7-\lambda & 12 \\ -1 & 0 & -1-\lambda \end{bmatrix} = 0 \\ &= (-5-\lambda)(7-\lambda)(-1-\lambda) + 4(7-\lambda) = 0 \\ &\Rightarrow (7-\lambda)(\lambda^2 + 6\lambda + 9) = 0 \end{aligned}$$

Thus the eigenvalues of A are $\lambda = 7$ and $\lambda = -3$

To check whether A is diagonalizable, we need to check the dimension of $\mathcal{N}(A + 3I)$

$$\begin{aligned}\mathcal{N}(A + 3I), & \begin{bmatrix} -2 & 0 & 4 & | & 0 \\ -11 & 10 & 12 & | & 0 \\ -1 & 0 & 2 & | & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & -2 & | & 0 \\ 0 & 1 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}\end{aligned}$$

Since there is only one linearly independent eigenvector, such as $\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$, associated with the eigenvalue of 3, we conclude that A is not diagonalizable.

To find a basis of $\mathbb{R}^3$ consisting of eigenvectors and generalised eigenvectors, we can look at the other eigenvectors associated with $\lambda = 7$ and pick a linearly independent vector from $\mathcal{N}(A - 7I)^2$

$$\begin{aligned}\mathcal{N}(A - 7I), & \begin{bmatrix} 12 & 0 & 4 & | & 0 \\ -11 & 0 & 12 & | & 0 \\ -1 & 0 & -8 & | & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}\end{aligned}$$

One such eigenvector is $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$$\begin{aligned}\mathcal{N}(A + 3I)^2, & \begin{bmatrix} 0 & 0 & 0 & | & 0 \\ -100 & 100 & 100 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 & | & 0 \\ -1 & 1 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}\end{aligned}$$

A linearly independent vector to the other two eigenvectors in this set is $\begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$

Hence the set $\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} \right\}$ is a basis for $\mathbb{R}^3$ made up of eigenvectors and a generalised eigenvector

Question 5

Assume for the sake of contradiction that $T - \lambda I$ is surjective,

Then $\text{Im}(T - \lambda I) = V$

Also, we know

$$\begin{aligned}\dim(\ker(T - \lambda I)) + \dim(\text{Im}(T - \lambda I)) &= \dim(V) \\ \Rightarrow \dim(\ker(T - \lambda I)) + \dim(V) &= \dim(V) \\ \Rightarrow \dim(\ker(T - \lambda I)) &= 0\end{aligned}$$

But since λ is an eigenvalue of T , we know $T - \lambda I$ has non-zero solutions, since eigenvectors are non-zero, this is a contradiction since if the $\dim(\ker T) = 0$ there exists a unique solution, the trivial one implying there are no non-zero eigenvectors.